Modelvragen hoofdstuk 4

Werner Peeters

1. Bereken door gebruik te maken van de regel van Fuss:
$$\int \frac{96x}{256 - x^4} dx$$
2. Bereken
$$\int \tan^5 x dx$$
3. Bereken door gebruik te maken van de regel van Fuss:
$$\int \frac{2x^2 - 3x + 4}{x^3 - 2x^2 + x} dx$$
4. Bereken
$$\int \frac{dx}{5 + 4\cos x}$$
5. Bereken door gebruik te maken van de regel van Fuss:
$$\int \frac{x^2 + x - 8}{x^3 + 4x} dx$$
6. Bereken
$$\int \sqrt{x^2 + 2x + 2} dx$$
7. Bereken
$$\int \frac{2x^4 + 7x^3 + 6x^2 - 5x - 1}{x^3 + 3x^2 - 4} dx$$
met de regel van Fuss.
8. Bereken
$$\int \frac{dx}{\sin x + \cos x + 2}$$
9. Bereken
$$\int e^{2x} \cos 4x dx$$
10. Bereken door gebruik te maken van de regel van Fuss:
$$\int \frac{3x + 1}{x^4 + x^3} dx$$

11. Bereken

$$\int \frac{1}{1+\sqrt[3]{x+1}}dx$$

12. Bereken door gebruik te maken van de regel van Fuss:

$$\int \frac{9x^2-4x-13}{x^3-7x+6}dx$$

13. Bereken

$$\int \frac{8x+8}{-x^3+4x^2-4x+16}dx$$

14. Bereken

$$\int \frac{dx}{\sqrt{6+x-x^2}}$$

15. Bereken

$$\int \frac{\sqrt{1-\ln x}}{x}dx$$

16. Bereken door gebruik te maken van de regel van Fuss:

$$\int \frac{4x^2+11x+12}{\left(x+2\right)^2\left(x-1\right)}dx$$

17. Bereken

$$\int \frac{dx}{3\cos x+4\sin x+5}$$

18. Bereken door gebruik te maken van de regel van Fuss:

$$\int \frac{3x^2-x+6}{\left(x^2+4\right)\left(x-2\right)}dx$$

Oplossingen:

1. Bereken door gebruik te maken van de regel van Fuss:

$$\int \frac{96x}{256-x^4}dx$$

$$\frac{96x}{256-x^4} = \frac{96x}{(4-x)(4+x)(16+x^2)} = \frac{A}{4-x} + \frac{B}{4+x} + \frac{Cx+D}{16+x^2}$$

$$\begin{aligned} A &= \left.\frac{96x}{(4+x)(16+x^2)}\right|_{x=4} = \frac{3}{2} \\ B &= \left.\frac{96x}{(4-x)(16+x^2)}\right|_{x=-4} = -\frac{3}{2} \\ C(4i)+D &= \left.\frac{96x}{(4-x)(4+x)}\right|_{x=4i} = 12i \\ &\Rightarrow \begin{cases} C=3 \\ D=0 \end{cases} \end{aligned}$$

$$\Rightarrow \int \frac{96x}{256-x^4}dx = \int \left(\frac{3}{2(4-x)} - \frac{3}{2(4+x)} + \frac{3x}{16+x^2}\right)dx = \frac{3}{2}\int \frac{dx}{4-x} - \frac{3}{2}\int \frac{dx}{4+x} + \frac{3}{2}\int \frac{d\left(x^2+16\right)}{x^2+16}$$
$$= -\frac{3}{2}\ln|4-x| - \frac{3}{2}\ln|4+x| + \frac{3}{2}\ln\left|16+x^2\right| + c = \frac{3}{2}\ln\left|\frac{16+x^2}{16-x^2}\right| + c$$

2. Bereken

$$\int \tan^5 x dx$$

$$\int \tan^5 x dx = \int \left(\tan^5 x + \tan^3 x\right)dx - \int \tan^3 x dx$$
$$= \int \tan^3 x \left(\tan^2 x + 1\right)dx - \int \tan^3 x dx$$
$$= \int \tan^3 x d(\tan x) - \int \left(\tan^3 x + \tan x\right)dx + \int \tan x dx$$
$$= \frac{1}{4}\tan^4 x - \int \tan x \left(\tan^2 x + 1\right)dx + \int \frac{\sin x}{\cos x}dx$$
$$= \frac{1}{4}\tan^4 x - \int \tan x d(\tan x) - \int \frac{d(\cos x)}{\cos x}$$
$$= \frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x - \ln|\cos x| + c$$

3. Bereken door gebruik te maken van de regel van Fuss:

$$\int \frac{2x^2-3x+4}{x^3-2x^2+x}dx$$

$$\frac{2x^2-3x+4}{x^3-2x^2+x} = \frac{2x^2-3x+4}{x(x-1)^2} = \frac{A}{x} + \frac{B}{(x-1)^2} + \frac{C}{x-1}$$

$$\begin{aligned}
A &= \left.\frac{2x^2-3x+4}{(x-1)^2}\right|_{x=0} = 4 \\
B &= \left.\frac{2x^2-3x+4}{x}\right|_{x=1} = 3 \\
&\frac{2x^2-3x+4}{x(x-1)^2} - \frac{3}{(x-1)^2} = \frac{2x^2-3x+4-3x}{(x-1)^2 x} = \frac{2(x-1)(x-2)}{(x-1)^2 x} = \frac{2(x-2)}{(x-1)x} \\
C &= \left.\frac{2(x-2)}{x}\right|_{x=1} = -2
\end{aligned}$$

$$\Rightarrow \int \frac{2x^2-3x+4}{x^3-2x^2+x}dx = \int \left(\frac{4}{x} + \frac{3}{(x-1)^2} - \frac{2}{x-1}\right)dx = 4\ln|x| - \frac{3}{x-1} - 2\ln|x-1| + c$$

4. Bereken

$$\int \frac{dx}{5+4\cos x}$$

$$\text{Stel } t = \tan\frac{x}{2} \Rightarrow \begin{cases} dx = \dfrac{2dt}{1+t^2} \\ \cos x = \dfrac{1-t^2}{1+t^2} \end{cases}$$

$$\begin{aligned}
&= \int \frac{\dfrac{2}{1+t^2}}{5+4\dfrac{1-t^2}{1+t^2}}dt \\
&= \int \frac{2}{5(1+t^2)+4(1-t^2)}dt \\
&= \int \frac{2}{5(1+t^2)+4(1-t^2)}dt \\
&= \int \frac{2dt}{t^2+9} \\
&= \frac{2}{3}\,\text{Bgtan}\left(\frac{1}{3}\tan\frac{x}{2}\right) + c
\end{aligned}$$

5. Bereken door gebruik te maken van de regel van Fuss:

$$\int \frac{x^2+x-8}{x^3+4x}dx$$

$$\frac{x^2+x-8}{x^3+4x} = \frac{x^2+x-8}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$$

$$\begin{aligned}
A &= \left.\frac{x^2+x-8}{(x^2+4)}\right|_{x=0} = -2 \\
B(2i)+C &= \left.\frac{x^2+x-8}{x}\right|_{x=2i} = 1+6i \\
&\Rightarrow \begin{cases} B = 3 \\ C = 1 \end{cases}
\end{aligned}$$

$$\Rightarrow \int \frac{x^2+x-8}{x^3+4x}dx = \int\left(-\frac{2}{x} + \frac{3x+1}{x^2+4}\right)dx = -2\ln|x| + \frac{3}{2}\int\frac{d\left(x^2+4\right)}{x^2+4} + \int\frac{dx}{x^2+4}$$
$$= -2\ln|x| + \frac{3}{2}\ln\left(x^2+4\right) + \frac{1}{2}\,\text{Bgtan}\,\frac{x}{2} + c$$

6. Bereken

$$\int \sqrt{x^2+2x+2}dx$$

$$\begin{aligned}
&\int \sqrt{x^2+2x+2}dx\\
&= \int \sqrt{(x+1)^2+1}dx\\
&\overset{y=x+1}{=} \int \sqrt{y^2+1}dy\\
&\overset{y=\operatorname{sh} t}{=} \int \sqrt{\operatorname{sh}^2 t+1}\operatorname{ch} t dt\\
&= \int \operatorname{ch}^2 t dt\\
&= \frac{1}{2}\int (1+\operatorname{ch} 2t)\, dt\\
&= \frac{t}{2}+\frac{\operatorname{sh} 2t}{4}+c\\
&= \frac{t}{2}+\frac{\operatorname{sh} t \operatorname{ch} t}{2}+c\\
&= \frac{t}{2}+\frac{\operatorname{sh} t\sqrt{1+\operatorname{sh}^2 t}}{2}+c\\
&= \frac{\operatorname{Bgsh} y}{2}+\frac{y\sqrt{1+y^2}}{2}+c\\
&= \frac{\operatorname{Bgsh}(x+1)}{2}+\frac{(x+1)\sqrt{x^2+2x+2}}{2}+c
\end{aligned}$$

Alternatief:

$\sqrt{x^2+2x+2} = x+t$

$\Rightarrow x^2+2x+2 = (x+t)^2 = x^2+2xt+t^2$

$$\begin{cases}
\Rightarrow x = \dfrac{t^2-2}{2-2t}\\
\Rightarrow dx = \dfrac{-t^2+2t-2}{2(t-1)^2}dt\\
\Rightarrow \sqrt{x^2+2x+2} = \dfrac{t^2-2}{2-2t}+t = \dfrac{t^2-2t+2}{2(t-1)}
\end{cases}$$

$$\begin{aligned}
&\Rightarrow I = \int \frac{t^2-2t+2}{2(t-1)}\frac{-t^2+2t-2}{2(t-1)^2}dt\\
&= \int \frac{t^2-2t+2}{2(t-1)}\frac{-t^2+2t-2}{2(t-1)^2}dt\\
&= \int \frac{t^4-4t^3+8t^2-8t+4}{4(1-t)^3}dt\\
&= \int \left(\frac{1}{4}-\frac{1}{4}t+\frac{2t^2-4t+3}{4(1-t)^3}\right)dt\\
&= \int \left(\frac{1}{4}-\frac{1}{4}t+\frac{1}{4(1-t)^3}-\frac{1}{2(t-1)}\right)dt\\
&= \frac{1}{4}t-\frac{1}{8}t^2+\frac{1}{8(1-t)^2}-\frac{1}{2}\ln|2t-2|+c
\end{aligned}$$

$= \frac{1}{4}\left(\sqrt{x^2+2x+2}-x\right) - \frac{1}{8}\left(\sqrt{x^2+2x+2}-x\right)^2 + \frac{1}{8\left(1-\left(\sqrt{x^2+2x+2}-x\right)\right)^2} - \frac{1}{2}\ln\left|2\left(\sqrt{x^2+2x+2}-x\right)-2\right| +$
c

7. Bereken
$$\int \frac{2x^4+7x^3+6x^2-5x-1}{x^3+3x^2-4}dx$$
met de regel van Fuss.
$\frac{2x^4+7x^3+6x^2-5x-1}{x^3+3x^2-4} = 2x+1+\frac{3x^2+3x+3}{x^3+3x^2-4}$
$\frac{3x^2+3x+3}{x^3+3x^2-4} = \frac{3+3x+3x^2}{(x-1)(x+2)^2} = \frac{A}{x-1}+\frac{B}{(x+2)^2}+\frac{C}{x+2}$
$$\begin{cases} A = \frac{3+3x+3x^2}{(x+2)^2}|_{x=1} = 1 \\ B = \frac{3+3x+3x^2}{(x-1)}|_{x=-2} = -3 \\ C \ = \left(\frac{3+3x+3x^2}{(x-1)(x+2)^2}+\frac{3}{(x+2)^2}\right)(x+2)\,|_{x=-2} \\ \quad = \left(\frac{3+3x+3x^2}{(x-1)(x+2)}+\frac{3}{(x+2)}\right)|_{x=-2} \\ \quad = \frac{3x}{x-1}|_{x=-2} = 2 \end{cases}$$
$\Rightarrow \frac{2x^4+7x^3+6x^2-5x-1}{x^3+3x^2-4} = 2x+1+\frac{1}{x-1}-\frac{3}{(x+2)^2}+\frac{2}{x+2}$
$\Rightarrow \int \frac{2x^4+7x^3+6x^2-5x-1}{x^3+3x^2-4}dx = x^2+x+\ln|x-1|+\frac{3}{x+2}+2\ln|x+2|+c$

8. Bereken
$$\int \frac{dx}{\sin x+\cos x+2}$$
Stel $t = \tan\frac{x}{2} \Rightarrow \begin{cases} \sin x = \frac{2t}{1+t^2} \\ \cos x = \frac{1-t^2}{1+t^2} \\ dx = \frac{2dt}{1+t^2} \end{cases}$
$\Rightarrow I = \int \frac{\frac{2dt}{1+t^2}}{\frac{2t}{1+t^2}+\frac{1-t^2}{1+t^2}+2} = \int \frac{2dt}{2t+1-t^2+2+2t^2} = \int \frac{2dt}{t^2+2t+3}$
$= \int \frac{2dt}{(t+1)^2+2} = \frac{1}{2}\int \frac{2dt}{\left(\frac{t+1}{\sqrt{2}}\right)^2+1} = \int \frac{dt}{\left(\frac{t+1}{\sqrt{2}}\right)^2+1}$
Stel $z = \frac{t+1}{\sqrt{2}} \Rightarrow dt = \sqrt{2}dz$
$I = \sqrt{2}\int \frac{dz}{z^2+1} = \sqrt{2}\,\mathrm{Bgtan}\,z + c = \sqrt{2}\,\mathrm{Bgtan}\,\frac{t+1}{\sqrt{2}} + c = \sqrt{2}\,\mathrm{Bgtan}\,\frac{\tan\frac{x}{2}+1}{\sqrt{2}} + c$

9. Bereken

$$\int e^{2x} \cos 4x dx$$

Stel $\left\{ \begin{array}{l} u = \cos 4x \\ dv = e^{2x} dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = -4 \sin 4x dx \\ v = \frac{1}{2} e^{2x} \end{array} \right.$

$\Rightarrow I = \frac{e^{2x} \cos 4x}{2} + 2 \int e^{2x} \sin 4x dx$

Stel $\left\{ \begin{array}{l} u = \sin 4x \\ dv = e^{2x} dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 4 \cos 4x dx \\ v = \frac{1}{2} e^{2x} \end{array} \right.$

$\Rightarrow I = \frac{e^{2x} \cos 4x}{2} + 2 \left(\frac{1}{2} e^{2x} \sin 4x - 2 \int e^{2x} \cos 4x dx \right)$

$\Rightarrow I = \frac{e^{2x} \cos 4x}{2} + e^{2x} \sin 4x - 4 \int e^{2x} \cos 4x dx$

$\Rightarrow 5I = \frac{e^{2x} \cos 4x}{2} + e^{2x} \sin 4x + c$

$\Rightarrow I = \frac{e^{2x} \cos 4x}{10} + \frac{e^{2x} \cos 4x}{5} + c$

10. Bereken door gebruik te maken van de regel van Fuss:

$$\int \frac{3x+1}{x^4+x^3} dx$$

$\int \frac{3x+1}{x^4+x^3} dx = \int \frac{3x+1}{x^3 (x+1)} dx$

Stel $\frac{3x+1}{x^4+x^3} = \frac{A}{x^3} + \frac{B}{x^2} + \frac{C}{x} + \frac{D}{x+1}$

$$\begin{array}{rcl} \Rightarrow A & = & \frac{3x+1}{x+1} |_{x=0} = 1 \\ & & \frac{3x+1}{x^3 (x+1)} - \frac{1}{x^3} = \frac{2}{x^2 (x+1)} \\ \Rightarrow B & = & B = \frac{2}{x+1} |_{x=0} = 2 \\ \Rightarrow C & = & -\frac{2}{x+1} |_{x=0} = -2 \\ \Rightarrow D & = & \frac{3x+1}{x^3} |_{x=-1} = 2 \end{array}$$

$\Rightarrow I = \int \frac{dx}{x^3} + \int \frac{2dx}{x^2} - \int \frac{2dx}{x} + \int \frac{2dx}{x+1} = -\frac{1}{2x^2} - \frac{2}{x} - 2 \ln |x| + 2 \ln |x+1| + c$

11. Bereken

$$\int \frac{1}{1+\sqrt[3]{x+1}} dx$$

Stel $t = 1 + \sqrt[3]{x+1} \Rightarrow x + 1 = (t-1)^3 = t^3 - 3t^2 + 3t - 1 \Rightarrow x = t^3 - 3t^2 + 3t - 2$

$dx = \left(3t^2 - 6t + 3\right) dt$

$\Rightarrow I = \int \frac{\left(3t^2 - 6t + 3\right)}{t} dt = \int \left(3t - 6 + \frac{3}{t}\right) dt = \frac{3}{2} t^2 - 6t + 3 \ln |t| + c$

$= \frac{3}{2} \left(1 + \sqrt[3]{x+1}\right)^2 - 6 \left(1 + \sqrt[3]{x+1}\right) + 3 \ln \left|1 + \sqrt[3]{x+1}\right| + c$

12. Bereken door gebruik te maken van de regel van Fuss:

$$\int \frac{9x^2-4x-13}{x^3-7x+6}dx$$

$$\int \frac{9x^2-4x-13}{x^3-7x+6}dx = \int \frac{9x^2-4x-13}{(x-1)(x-2)(x+3)}dx$$

Stel $\frac{9x^2-4x-13}{(x-1)(x-2)(x+3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x+3}$

$$\Rightarrow \begin{cases} A = \frac{9x^2-4x-13}{(x-2)(x+3)}|_{x=1} = 2 \\ B = \frac{9x^2-4x-13}{(x-1)(x+3)}|_{x=2} = 3 \\ C = \frac{9x^2-4x-13}{(x-1)(x-2)}|_{x=-3} = 4 \end{cases}$$

$$\Rightarrow I = \int \frac{2dx}{x-1} + \int \frac{3dx}{x-2} + \int \frac{4dx}{x+3} = 2\ln|x-1| + 3\ln|x-2| + 4\ln|x+3| + c$$

13. Bereken

$$\int \frac{8x+8}{-x^3+4x^2-4x+16}dx$$

$$\frac{8x+8}{-x^3+4x^2-4x+16} = \frac{-8x-8}{(x-4)(x^2+4)} = \frac{A}{x-4} + \frac{Bx+C}{x^2+4}$$

$$\begin{aligned} A &= \lim_{x\to 4} \frac{-8x-8}{(x^2+4)} = -2 \\ 2Bi+C &= \lim_{x\to 2i} \frac{-8x-8}{x-4} = 4i \Rightarrow \begin{cases} B = 2 \\ C = 0 \end{cases} \end{aligned}$$

$$\Rightarrow I = \int \frac{-2dx}{x-4} + \int \frac{2xdx}{x^2+4} = \ln\left(x^2+4\right) - 2\ln|x-4| + c$$

14. Bereken

$$\int \frac{dx}{\sqrt{6+x-x^2}}$$

$$\int \frac{dx}{\sqrt{6+x-x^2}} = \int \frac{dx}{\sqrt{(3-x)(x+2)}}$$

Stel

$$\sqrt{(3-x)(x+2)} = (3-x)\,t,$$

dan is

$$x+2 = (3-x)\,t^2$$

en dus

$$x = \frac{3t^2-2}{t^2+1}.$$

Daaruit volgt dat

$$3-x = \frac{5}{t^2+1}$$

en dat

$$dx = \frac{10t dt}{\left(t^2+1\right)^2}.$$

Substitutie geeft ons dan

$$I = \int \frac{\dfrac{10t dt}{\left(t^2+1\right)^2}}{\dfrac{5t}{t^2+1}} = \int \frac{2dt}{t^2+1} = 2\,\mathrm{Bgtan}\, t + k = 2\,\mathrm{Bgtan}\sqrt{\frac{x+2}{3-x}} + k$$

Of: $\displaystyle\int \frac{dx}{\sqrt{6+x-x^2}} = \int \frac{dx}{\sqrt{\frac{25}{4}-\left(x-\frac{1}{2}\right)^2}} = \frac{2}{5}\int \frac{dx}{\sqrt{1-\left(\frac{2x-1}{5}\right)^2}}$

$$= \int \frac{d\left(\frac{2x-1}{5}\right)}{\sqrt{1-\left(\frac{2x-1}{5}\right)^2}} = \mathrm{Bgsin}\left(\frac{2}{5}x - \frac{1}{5}\right) + c$$

15. Bereken

$$\int \frac{\sqrt{1-\ln x}}{x} dx$$

Stel $t = 1 - \ln x \Rightarrow dt = -\dfrac{dx}{x} \Rightarrow I = -\displaystyle\int \sqrt{t} dt = -\frac{2}{3} t^{\frac{3}{2}} + c = -\frac{2}{3}\sqrt{\left(1-\ln x\right)^3} + c$

16. Bereken door gebruik te maken van de regel van Fuss:

$$\int \frac{4x^2+11x+12}{\left(x+2\right)^2\left(x-1\right)} dx$$

Stel $\dfrac{4x^2+11x+12}{\left(x+2\right)^2\left(x-1\right)} = \dfrac{A}{\left(x+2\right)^2} + \dfrac{B}{\left(x+2\right)} + \dfrac{C}{x-1}$

$$\begin{aligned}
\Rightarrow A &= \frac{4x^2+11x+12}{x-1}\Big|_{x=-2} = -2 \\
& \frac{4x^2+11x+12}{\left(x+2\right)^2\left(x-1\right)} + \frac{2}{\left(x+2\right)^2} = \frac{4x^2+11x+12+2\left(x-1\right)}{\left(x+2\right)^2\left(x-1\right)} = \frac{4x+5}{\left(x+2\right)\left(x-1\right)} \\
\Rightarrow B &= \frac{4x+5}{\left(x-1\right)}\Big|_{x=-2} = 1 \\
\Rightarrow C &= \frac{4x^2+11x+12}{\left(x+2\right)^2}\Big|_{x=1} = 3
\end{aligned}$$

$$I = \int \left(\frac{1}{x+2} - \frac{2}{\left(x+2\right)^2} + \frac{3}{x-1}\right) dx = \frac{2}{x+2} + \ln|x+2| + 3\ln|x-1| + c$$

17. Bereken

$$\int \frac{dx}{3\cos x + 4\sin x + 5}$$

Stel $t = \tan\dfrac{x}{2}$

$$\Rightarrow I = \int \frac{\frac{2dt}{1+t^2}}{3\frac{1-t^2}{1+t^2} + 4\frac{2t}{1+t^2} + 5} = \int \frac{2dt}{2t^2+8t+8} = \int \frac{dt}{t^2+4t+4}$$

$$= \int \frac{dt}{(t+2)^2} = -\frac{1}{t+2} + c = -\frac{1}{\tan\frac{x}{2}+2} + c$$

18. Bereken door gebruik te maken van de regel van Fuss:

$$\int \frac{3x^2 - x + 6}{(x^2+4)(x-2)} dx$$

Stel $\dfrac{3x^2 - x + 6}{(x^2+4)(x-2)} = \dfrac{Ax+B}{x^2+4} + \dfrac{C}{x-2}$

$$\Rightarrow \quad A(2i) + B = \frac{3x^2 - x + 6}{x-2}\Big|_{x=2i} = 1 + 2i \Rightarrow \begin{cases} A = 1 \\ B = 1 \end{cases}$$

$$\Rightarrow \quad C = \frac{3x^2 - x + 6}{x^2+4}\Big|_{x=2} = 2$$

$$\Rightarrow I = \int \left(\frac{x+1}{x^2+4} + \frac{2}{x-2} \right) dx = \frac{1}{2}\ln\left|x^2+4\right| + \frac{1}{2}\,\mathrm{Bgtan}\,\frac{x}{2} + 2\ln|x-2| + c$$